

ICT731 Data Mining

Term 2, 2025

Week 7 Tutorial

(Submit this tutorial to Moodle by **Friday 22/08/2025 @ 23.59**)

Name: Sumit Kafle

Student ID: 20032965

Subject Name: Data Mining

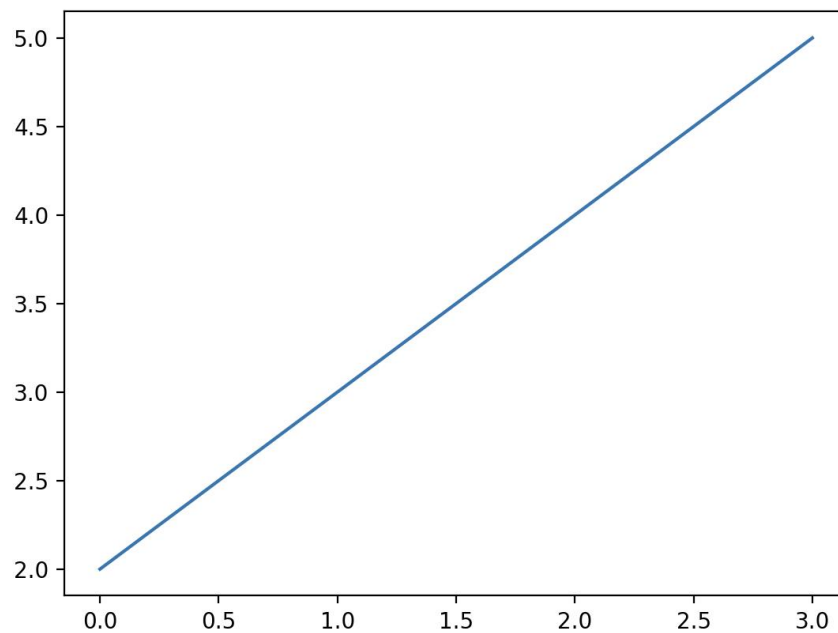
1. Implement and run the **Python regression programs** discussed in **Week 7 Lecture** saving **Python screen captures** of your operations and submitting them.
2. Install **RapidMiner** and go through **Textbook** pages **972-984** (for the **online version of the textbook** see pages **522-526**) saving **RapidMiner screen captures** of your operations and submitting them.

Question 1(Solution)

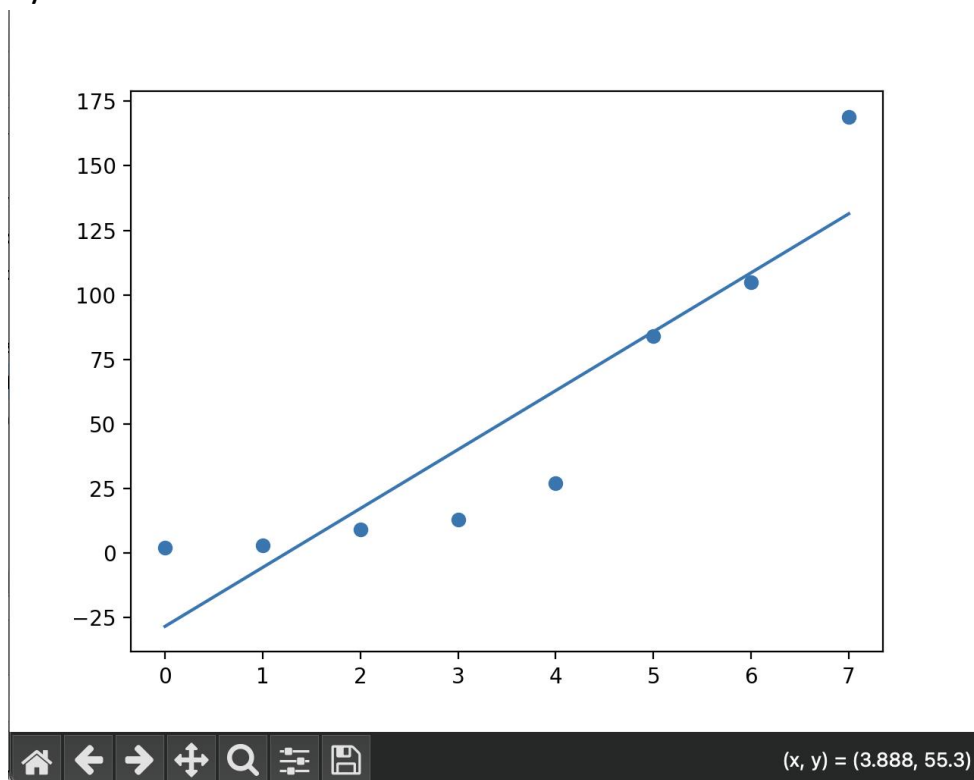
Command showing SimpleLR and MultipleLR:

```
sumitkafle@Sumits-MacBook-Air Lecture7Files % source ~/datamining/.venv/bin/activate  
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 SimpleLR.py  
[Coefficient: [1.]  
Y intercept: 2.0  
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 MultipleLR.py  
  
Coefficient: [-3.22147536e-15 3.00000000e+00 3.00000000e+00]  
Y intercept: -5.0000000000000014  
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 Plot.py  
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files %
```

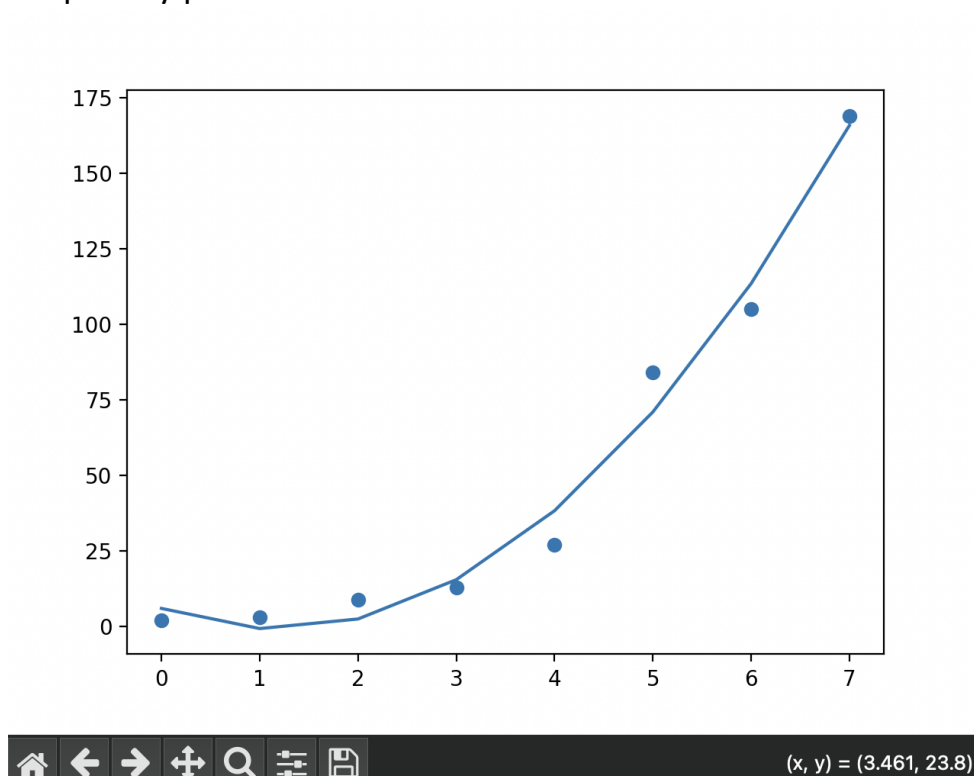
Plotpy:



PythonLR:



SimplePoly plot:



SimpleDecisionTreeRegression cmd

```
[(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 SimpleDecisionTreeRegression.py  
[1.]  
Actual: 2 Predicted: 2.0  
Actual: 3 Predicted: 3.0  
Actual: 4 Predicted: 4.0  
Actual: 5 Predicted: 5.0  
PDF created  
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files %
```

WeightMPG:

```
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 WeightMPG.py
Coefficient: [0.00669058]
Actual: 18.0 Predicted: 23.443794170864088 Weight: 3504
Actual: 15.0 Predicted: 24.708313890696655 Weight: 3693
Actual: 18.0 Predicted: 22.988834694945492 Weight: 3436
Actual: 16.0 Predicted: 22.968762953360848 Weight: 3433
Actual: 17.0 Predicted: 23.075812241812283 Weight: 3449
Actual: 15.0 Predicted: 29.04381007297974 Weight: 4341
Actual: 14.0 Predicted: 29.13078761984653 Weight: 4354
Actual: 14.0 Predicted: 28.849783237661512 Weight: 4312
Actual: 14.0 Predicted: 29.60581883734977 Weight: 4425
Actual: 15.0 Predicted: 25.75873503362635 Weight: 3850
Actual: 15.0 Predicted: 23.83853842202875 Weight: 3563
Actual: 14.0 Predicted: 24.146305126326624 Weight: 3609
Actual: 15.0 Predicted: 25.16327336661525 Weight: 3761
Actual: 14.0 Predicted: 20.64713151007037 Weight: 3086
Actual: 24.0 Predicted: 15.870057012925118 Weight: 2372
Actual: 22.0 Predicted: 18.954414636432066 Weight: 2833
Actual: 18.0 Predicted: 18.559670385267403 Weight: 2774
Actual: 21.0 Predicted: 17.308531826491265 Weight: 2587
Actual: 27.0 Predicted: 14.250936525097176 Weight: 2130
Actual: 26.0 Predicted: 12.277215269273858 Weight: 1835
Actual: 25.0 Predicted: 17.87723117138951 Weight: 2672
Actual: 24.0 Predicted: 16.25811068356157 Weight: 2430
Actual: 25.0 Predicted: 15.890128754509762 Weight: 2375
Actual: 26.0 Predicted: 14.946756900031499 Weight: 2234
Actual: 21.0 Predicted: 17.716657238712358 Weight: 2648
Actual: 10.0 Predicted: 30.877029137710547 Weight: 4615
Actual: 10.0 Predicted: 29.27798039146725 Weight: 4376
Actual: 11.0 Predicted: 29.318123874636537 Weight: 4382
Actual: 9.0 Predicted: 31.65982705951166 Weight: 4732
Actual: 27.0 Predicted: 14.250936525097176 Weight: 2130
Actual: 28.0 Predicted: 15.147474315877938 Weight: 2264
Actual: 25.0 Predicted: 14.90661341686221 Weight: 2228
Actual: 19.0 Predicted: 17.622989111317352 Weight: 2634
Actual: 16.0 Predicted: 23.008906436530136 Weight: 3439
Actual: 17.0 Predicted: 22.272942578426527 Weight: 3329
Actual: 19.0 Predicted: 22.09229690416473 Weight: 3302
Actual: 18.0 Predicted: 21.998628776769724 Weight: 3288
Actual: 14.0 Predicted: 28.160653443255406 Weight: 4209
Actual: 14.0 Predicted: 29.86675147795014 Weight: 4464
Actual: 14.0 Predicted: 27.7926715142036 Weight: 4154
Actual: 14.0 Predicted: 27.404617843567152 Weight: 4096
Actual: 12.0 Predicted: 33.151826517303526 Weight: 4955
Actual: 13.0 Predicted: 31.753495186906665 Weight: 4746
Actual: 13.0 Predicted: 34.38958391502323 Weight: 5140
Actual: 18.0 Predicted: 19.817499524571755 Weight: 2962
Actual: 22.0 Predicted: 16.110917911940845 Weight: 2408
Actual: 19.0 Predicted: 21.95848529360044 Weight: 3282
Actual: 18.0 Predicted: 21.001732278065745 Weight: 3139
Actual: 23.0 Predicted: 14.853088772636493 Weight: 2220
Actual: 28.0 Predicted: 14.204102461399675 Weight: 2123
Actual: 30.0 Predicted: 13.876264015517156 Weight: 2074
Actual: 30.0 Predicted: 13.816048790763224 Weight: 2065
Actual: 31.0 Predicted: 11.86239927652455 Weight: 1773
```

AutoMPGMR:

```
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7/Files % python3 AutoMPGMR.py
Coefficient: [-0.00607987 -0.03489977 -0.62739129 -0.06569545  0.01930388  0.5804971
1.10007702]
Actual: 18.0 Predicted: 16.012841738633124 Weight: 3504.0
Actual: 15.0 Predicted: 14.505168381684932 Weight: 3693.0
Actual: 18.0 Predicted: 16.006316011283474 Weight: 3436.0
Actual: 16.0 Predicted: 15.688605843780262 Weight: 3433.0
Actual: 17.0 Predicted: 16.00026092119312 Weight: 3449.0
Actual: 15.0 Predicted: 11.037267543836645 Weight: 4341.0
Actual: 14.0 Predicted: 10.738726828451515 Weight: 4354.0
Actual: 14.0 Predicted: 10.931173747974277 Weight: 4312.0
Actual: 14.0 Predicted: 10.086165376643038 Weight: 4425.0
Actual: 15.0 Predicted: 13.647375623237867 Weight: 3850.0
Actual: 15.0 Predicted: 15.856624446673301 Weight: 3563.0
Actual: 14.0 Predicted: 15.22727188082484 Weight: 3609.0
Actual: 15.0 Predicted: 15.711818401075117 Weight: 3761.0
Actual: 14.0 Predicted: 18.227116751246648 Weight: 3086.0
Actual: 24.0 Predicted: 24.88443079103429 Weight: 2372.0
Actual: 22.0 Predicted: 20.23465444742926 Weight: 2833.0
Actual: 18.0 Predicted: 20.542871368040046 Weight: 2774.0
Actual: 21.0 Predicted: 22.085061161526944 Weight: 2587.0
Actual: 27.0 Predicted: 26.324044291498694 Weight: 2130.0
Actual: 26.0 Predicted: 28.089147588978097 Weight: 1835.0
Actual: 25.0 Predicted: 22.017439415678496 Weight: 2672.0
Actual: 24.0 Predicted: 23.523244341109105 Weight: 2430.0
Actual: 25.0 Predicted: 23.428140600298303 Weight: 2375.0
Actual: 26.0 Predicted: 24.313850290166386 Weight: 2234.0
Actual: 21.0 Predicted: 21.586081582204237 Weight: 2648.0
Actual: 10.0 Predicted: 7.1833364178068955 Weight: 4615.0
Actual: 10.0 Predicted: 8.07112162966175 Weight: 4376.0
Actual: 11.0 Predicted: 7.996530599560959 Weight: 4382.0
Actual: 9.0 Predicted: 5.863139109181315 Weight: 4732.0
Actual: 27.0 Predicted: 26.904541391465873 Weight: 2130.0
Actual: 28.0 Predicted: 24.58425616538144 Weight: 2264.0
Actual: 25.0 Predicted: 26.40612520402817 Weight: 2228.0
Actual: 19.0 Predicted: 22.67111825203462 Weight: 2634.0
Actual: 16.0 Predicted: 17.302955011721437 Weight: 3439.0
Actual: 17.0 Predicted: 18.62883702584807 Weight: 3329.0
Actual: 19.0 Predicted: 19.211790810910557 Weight: 3302.0
Actual: 18.0 Predicted: 18.53064199401346 Weight: 3288.0
Actual: 14.0 Predicted: 11.915602739758638 Weight: 4209.0
Actual: 14.0 Predicted: 11.014279001536368 Weight: 4464.0
Actual: 14.0 Predicted: 12.589553709012899 Weight: 4154.0
Actual: 14.0 Predicted: 12.44270533258147 Weight: 4096.0
Actual: 12.0 Predicted: 7.52639602088624 Weight: 4955.0
Actual: 13.0 Predicted: 9.441405618325286 Weight: 4746.0
Actual: 13.0 Predicted: 6.87143640905545 Weight: 5140.0
Actual: 18.0 Predicted: 20.796975101420305 Weight: 2962.0
Actual: 22.0 Predicted: 24.10701601152351 Weight: 2408.0
Actual: 19.0 Predicted: 18.947438830707128 Weight: 3282.0
Actual: 18.0 Predicted: 20.268505731194512 Weight: 3139.0
Actual: 23.0 Predicted: 24.74244298875097 Weight: 2220.0
Actual: 28.0 Predicted: 26.176845437413746 Weight: 2123.0
Actual: 30.0 Predicted: 26.097185995528953 Weight: 2074.0
Actual: 30.0 Predicted: 26.444718441250416 Weight: 2065.0
Actual: 31.0 Predicted: 29.080220603224642 Weight: 1773.0
Actual: 35.0 Predicted: 29.998400721137443 Weight: 1613.0
Actual: 27.0 Predicted: 28.285671018103294 Weight: 1834.0
Actual: 26.0 Predicted: 25.886565125418464 Weight: 1955.0
Actual: 24.0 Predicted: 26.58408542943977 Weight: 2278.0
Actual: 25.0 Predicted: 25.433815402225672 Weight: 2126.0
Actual: 23.0 Predicted: 26.226390089175357 Weight: 2254.0
```

AutoMPGRF:

```
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 AutoMPGRF.py
```

```
[0.22213432 0.13498909 0.13615394 0.0329148 0.35001293 0.11571375  
0.00808117]
```

```
Actual: 18.0 Predicted: 17.566999999999997 Weight: 3504.0  
Actual: 15.0 Predicted: 14.97 Weight: 3693.0  
Actual: 18.0 Predicted: 16.77 Weight: 3436.0  
Actual: 16.0 Predicted: 16.302 Weight: 3433.0  
Actual: 17.0 Predicted: 16.924 Weight: 3449.0  
Actual: 15.0 Predicted: 14.54 Weight: 4341.0  
Actual: 14.0 Predicted: 14.01 Weight: 4354.0  
Actual: 14.0 Predicted: 14.175 Weight: 4312.0  
Actual: 14.0 Predicted: 13.66 Weight: 4425.0  
Actual: 15.0 Predicted: 14.8 Weight: 3850.0  
Actual: 15.0 Predicted: 15.14 Weight: 3563.0  
Actual: 14.0 Predicted: 14.565 Weight: 3609.0  
Actual: 15.0 Predicted: 15.12 Weight: 3761.0  
Actual: 14.0 Predicted: 14.42 Weight: 3086.0  
Actual: 24.0 Predicted: 23.962999999999997 Weight: 2372.0  
Actual: 22.0 Predicted: 21.633000000000003 Weight: 2833.0  
Actual: 18.0 Predicted: 19.025 Weight: 2774.0  
Actual: 21.0 Predicted: 21.105 Weight: 2587.0  
Actual: 27.0 Predicted: 26.9 Weight: 2130.0  
Actual: 26.0 Predicted: 26.611 Weight: 1835.0  
Actual: 25.0 Predicted: 24.69 Weight: 2672.0  
Actual: 24.0 Predicted: 24.459 Weight: 2430.0  
Actual: 25.0 Predicted: 24.839000000000002 Weight: 2375.0  
Actual: 26.0 Predicted: 24.951999999999998 Weight: 2234.0  
Actual: 21.0 Predicted: 21.052 Weight: 2648.0  
Actual: 10.0 Predicted: 10.66 Weight: 4615.0  
Actual: 10.0 Predicted: 11.135 Weight: 4376.0  
Actual: 11.0 Predicted: 11.61 Weight: 4382.0  
Actual: 9.0 Predicted: 10.08 Weight: 4732.0  
Actual: 27.0 Predicted: 26.96 Weight: 2130.0  
Actual: 28.0 Predicted: 26.229 Weight: 2264.0  
Actual: 25.0 Predicted: 24.935 Weight: 2228.0  
Actual: 19.0 Predicted: 19.563 Weight: 2634.0  
Actual: 16.0 Predicted: 16.81 Weight: 3439.0  
Actual: 17.0 Predicted: 17.55 Weight: 3329.0  
Actual: 19.0 Predicted: 18.548 Weight: 3302.0  
Actual: 18.0 Predicted: 18.224 Weight: 3288.0  
Actual: 14.0 Predicted: 13.925 Weight: 4209.0  
Actual: 14.0 Predicted: 13.62 Weight: 4464.0  
Actual: 14.0 Predicted: 14.075 Weight: 4154.0  
Actual: 14.0 Predicted: 14.17 Weight: 4096.0  
Actual: 12.0 Predicted: 12.11 Weight: 4955.0  
Actual: 13.0 Predicted: 12.89 Weight: 4746.0  
Actual: 13.0 Predicted: 12.74 Weight: 5140.0  
Actual: 18.0 Predicted: 18.475 Weight: 2962.0  
Actual: 22.0 Predicted: 22.119 Weight: 2408.0
```


KNNRegression:

```
((.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 KNNRegression.py
Actual: 18.0 Predicted: 18.240000000000002 Weight: 3504.0
Actual: 15.0 Predicted: 15.280000000000001 Weight: 3693.0
Actual: 18.0 Predicted: 17.04 Weight: 3436.0
Actual: 16.0 Predicted: 17.04 Weight: 3433.0
Actual: 17.0 Predicted: 17.04 Weight: 3449.0
Actual: 15.0 Predicted: 14.5 Weight: 4341.0
Actual: 14.0 Predicted: 14.5 Weight: 4354.0
Actual: 14.0 Predicted: 14.9 Weight: 4312.0
Actual: 14.0 Predicted: 13.8 Weight: 4425.0
Actual: 15.0 Predicted: 16.84 Weight: 3850.0
Actual: 15.0 Predicted: 15.64 Weight: 3563.0
Actual: 14.0 Predicted: 15.440000000000001 Weight: 3609.0
Actual: 15.0 Predicted: 16.52 Weight: 3761.0
Actual: 14.0 Predicted: 20.22 Weight: 3086.0
Actual: 24.0 Predicted: 23.8 Weight: 2372.0
Actual: 22.0 Predicted: 23.54 Weight: 2833.0
Actual: 18.0 Predicted: 22.119999999999997 Weight: 2774.0
Actual: 21.0 Predicted: 24.119999999999997 Weight: 2587.0
Actual: 27.0 Predicted: 27.0 Weight: 2130.0
Actual: 26.0 Predicted: 29.660000000000004 Weight: 1835.0
Actual: 25.0 Predicted: 24.880000000000003 Weight: 2672.0
Actual: 24.0 Predicted: 27.78 Weight: 2430.0
Actual: 25.0 Predicted: 26.2 Weight: 2375.0
Actual: 26.0 Predicted: 24.0 Weight: 2234.0
Actual: 21.0 Predicted: 24.159999999999997 Weight: 2648.0
Actual: 10.0 Predicted: 12.8 Weight: 4615.0
Actual: 10.0 Predicted: 13.48 Weight: 4376.0
Actual: 11.0 Predicted: 13.48 Weight: 4382.0
Actual: 9.0 Predicted: 12.4 Weight: 4732.0
Actual: 27.0 Predicted: 27.0 Weight: 2130.0
Actual: 28.0 Predicted: 28.439999999999998 Weight: 2264.0
Actual: 25.0 Predicted: 24.0 Weight: 2228.0
Actual: 19.0 Predicted: 23.119999999999997 Weight: 2634.0
Actual: 16.0 Predicted: 19.48 Weight: 3439.0
Actual: 17.0 Predicted: 18.580000000000002 Weight: 3329.0
Actual: 19.0 Predicted: 17.8 Weight: 3302.0
Actual: 18.0 Predicted: 18.2 Weight: 3288.0
Actual: 14.0 Predicted: 14.8 Weight: 4209.0
Actual: 14.0 Predicted: 12.8 Weight: 4464.0
Actual: 14.0 Predicted: 14.8 Weight: 4154.0
Actual: 14.0 Predicted: 14.7 Weight: 4096.0
Actual: 12.0 Predicted: 11.8 Weight: 4955.0
Actual: 13.0 Predicted: 12.8 Weight: 4746.0
Actual: 13.0 Predicted: 12.0 Weight: 5140.0
Actual: 18.0 Predicted: 22.4 Weight: 2962.0
Actual: 22.0 Predicted: 23.0 Weight: 2408.0
Actual: 19.0 Predicted: 18.44 Weight: 3282.0
Actual: 18.0 Predicted: 17.8 Weight: 3139.0
Actual: 23.0 Predicted: 25.54 Weight: 2220.0
Actual: 28.0 Predicted: 27.82 Weight: 2123.0
Actual: 30.0 Predicted: 33.04 Weight: 2074.0
Actual: 30.0 Predicted: 33.04 Weight: 2065.0
Actual: 31.0 Predicted: 34.239999999999995 Weight: 1773.0
Actual: 35.0 Predicted: 34.239999999999995 Weight: 1613.0
Actual: 27.0 Predicted: 29.660000000000004 Weight: 1834.0
Actual: 26.0 Predicted: 29.7 Weight: 1955.0
Actual: 24.0 Predicted: 25.04 Weight: 2278.0
Actual: 25.0 Predicted: 28.6 Weight: 2126.0
Actual: 23.0 Predicted: 28.24 Weight: 2254.0
Actual: 20.0 Predicted: 21.78 Weight: 2408.0
```


SettleHousing:

```
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 SeattleHousing.py

Coefficient: [-3.55429634e+04  4.10775605e+04  1.10335560e+02  1.33179788e-01
 5.27005384e+03  5.83351196e+05  5.24194027e+04  2.72350677e+04
 9.55767038e+04  7.14334350e+01  3.89021250e+01 -2.56410126e+03
 2.04290662e+01 -5.19355977e+02  6.02267189e+05 -2.18124673e+05
 2.30910133e+01 -3.75943742e-01]

Actual: 221900.0 Predicted: 214074.58212679997
Actual: 538000.0 Predicted: 736220.0183606789
Actual: 180000.0 Predicted: 376666.31185634603
Actual: 604000.0 Predicted: 459571.47462562326
Actual: 510000.0 Predicted: 441180.97865586967
Actual: 1230000.0 Predicted: 1460407.4176285644
Actual: 257500.0 Predicted: 270625.07172242104
Actual: 291850.0 Predicted: 139474.16106344428
Actual: 229500.0 Predicted: 330775.9589167788
Actual: 323000.0 Predicted: 247994.95412902613
Actual: 662500.0 Predicted: 860626.5453870407
Actual: 468000.0 Predicted: 444514.4426061092
Actual: 310000.0 Predicted: 589700.9361844352
Actual: 400000.0 Predicted: 318660.3054068279
Actual: 530000.0 Predicted: 605294.6941679522
Actual: 650000.0 Predicted: 951013.9864668166
Actual: 395000.0 Predicted: 448771.44014180487
Actual: 485000.0 Predicted: 541525.6055084158
Actual: 189000.0 Predicted: 321286.09607226093
Actual: 230000.0 Predicted: 211131.1537276907
Actual: 385000.0 Predicted: 470797.5462333348
Actual: 2000000.0 Predicted: 1112329.7836171791
Actual: 285000.0 Predicted: 335666.10227741074
Actual: 252700.0 Predicted: 131888.33196274194
Actual: 329000.0 Predicted: 542574.4484232179
Actual: 233000.0 Predicted: 300577.5602681327
Actual: 937000.0 Predicted: 796290.9321273589
Actual: 667000.0 Predicted: 650334.2987974862
Actual: 438000.0 Predicted: 458586.7351238028
Actual: 719000.0 Predicted: 646776.5636328586
Actual: 580500.0 Predicted: 544152.4393913357
Actual: 280000.0 Predicted: 331737.1950781712
Actual: 687500.0 Predicted: 650104.3735125929
Actual: 535000.0 Predicted: 559443.4037416963
Actual: 322500.0 Predicted: 323175.8433681407
Actual: 696000.0 Predicted: 810679.3910521124
Actual: 550000.0 Predicted: 202511.30189992962
Actual: 640000.0 Predicted: 857894.6851280547
Actual: 240000.0 Predicted: 101012.29767922578
Actual: 605000.0 Predicted: 583028.9018085401
Actual: 625000.0 Predicted: 665972.7028175985
Actual: 775000.0 Predicted: 754756.5797892736
Actual: 861990.0 Predicted: 868658.6660177031
Actual: 685000.0 Predicted: 513410.80728769273
Actual: 309000.0 Predicted: 147962.79808344974
Actual: 488000.0 Predicted: 814452.7840709331
Actual: 210490.0 Predicted: 34497.015222730945
Actual: 785000.0 Predicted: 758712.9304646383
Actual: 450000.0 Predicted: 425413.3384680437
Actual: 1350000.0 Predicted: 1529040.821310391
Actual: 228000.0 Predicted: 189830.18520905555
Actual: 345000.0 Predicted: 660146.6353768616
Actual: 600000.0 Predicted: 476931.93174290744
Actual: 585000.0 Predicted: 499334.447598177
Actual: 920000.0 Predicted: 789595.6299734409
Actual: 885000.0 Predicted: 782717.3043077829
Actual: 292500.0 Predicted: 277476.87453564804
Actual: 301000.0 Predicted: 464162.7952177647
```

SettleDT:

```
Actual: 0.000000 Predicted: 0.000000  
(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 SeattleDT.py  
[0.00224554 0.00450036 0.25360808 0.01921312 0.00097777 0.03342615  
 0.00763779 0.0022107 0.33889434 0.02859965 0.0047575 0.01726669  
 0.00125904 0.01269038 0.16589315 0.06444247 0.02714218 0.0152351 ]  
Actual: 221900.0 Predicted: 221900.0  
Actual: 538000.0 Predicted: 538000.0  
Actual: 180000.0 Predicted: 180000.0  
Actual: 604000.0 Predicted: 604000.0  
Actual: 510000.0 Predicted: 510000.0  
Actual: 1230000.0 Predicted: 1230000.0  
Actual: 257500.0 Predicted: 257500.0  
Actual: 291850.0 Predicted: 291850.0  
Actual: 229500.0 Predicted: 229500.0  
Actual: 323000.0 Predicted: 323000.0  
Actual: 662500.0 Predicted: 662500.0  
Actual: 468000.0 Predicted: 468000.0  
Actual: 310000.0 Predicted: 310000.0  
Actual: 400000.0 Predicted: 400000.0  
Actual: 530000.0 Predicted: 530000.0  
Actual: 650000.0 Predicted: 650000.0  
Actual: 395000.0 Predicted: 395000.0  
Actual: 485000.0 Predicted: 485000.0  
Actual: 189000.0 Predicted: 189000.0  
Actual: 230000.0 Predicted: 230000.0  
Actual: 385000.0 Predicted: 385000.0  
Actual: 2000000.0 Predicted: 2000000.0  
Actual: 285000.0 Predicted: 285000.0  
Actual: 252700.0 Predicted: 252700.0  
Actual: 329000.0 Predicted: 329000.0  
Actual: 233000.0 Predicted: 233000.0  
Actual: 937000.0 Predicted: 937000.0  
Actual: 667000.0 Predicted: 667000.0  
Actual: 438000.0 Predicted: 438000.0  
Actual: 719000.0 Predicted: 719000.0  
Actual: 580500.0 Predicted: 580500.0  
Actual: 280000.0 Predicted: 280000.0  
Actual: 687500.0 Predicted: 687500.0  
Actual: 535000.0 Predicted: 535000.0  
Actual: 322500.0 Predicted: 322500.0  
Actual: 696000.0 Predicted: 696000.0  
Actual: 550000.0 Predicted: 550000.0  
Actual: 640000.0 Predicted: 640000.0  
Actual: 240000.0 Predicted: 240000.0  
Actual: 605000.0 Predicted: 605000.0  
Actual: 625000.0 Predicted: 625000.0  
Actual: 775000.0 Predicted: 775000.0  
Actual: 861990.0 Predicted: 861990.0  
Actual: 685000.0 Predicted: 685000.0  
Actual: 309000.0 Predicted: 309000.0  
Actual: 488000.0 Predicted: 488000.0  
Actual: 210490.0 Predicted: 210490.0  
Actual: 785000.0 Predicted: 785000.0  
Actual: 450000.0 Predicted: 450000.0  
Actual: 1350000.0 Predicted: 1350000.0  
Actual: 228000.0 Predicted: 228000.0  
Actual: 345000.0 Predicted: 345000.0  
Actual: 600000.0 Predicted: 600000.0  
Actual: 585000.0 Predicted: 585000.0  
Actual: 920000.0 Predicted: 920000.0  
Actual: 885000.0 Predicted: 885000.0  
Actual: 292500.0 Predicted: 292500.0  
Actual: 301000.0 Predicted: 301000.0  
Actual: 951000.0 Predicted: 951000.0  
Actual: 430000.0 Predicted: 430000.0
```

SettlePoly:

```
Actual: 325000.0 Predicted: 325000.0  
[(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 SeattlePoly.py  
Actual: 221900.0 Predicted: 281070.8905628808  
Actual: 538000.0 Predicted: 644960.2126952298  
Actual: 180000.0 Predicted: 533200.35893308  
Actual: 604000.0 Predicted: 373337.22232018784  
Actual: 510000.0 Predicted: 445433.7279790528  
Actual: 1230000.0 Predicted: 1601730.009312261  
Actual: 257500.0 Predicted: 215796.74003611878  
Actual: 291850.0 Predicted: 249890.6248906739  
Actual: 229500.0 Predicted: 330189.6368940957  
Actual: 323000.0 Predicted: 284981.73922740296  
Actual: 662500.0 Predicted: 690147.6714831479  
Actual: 468000.0 Predicted: 404936.53420316055  
Actual: 310000.0 Predicted: 593073.1352954991  
Actual: 400000.0 Predicted: 352239.7134328969  
Actual: 530000.0 Predicted: 679166.9644778855  
Actual: 650000.0 Predicted: 871897.6550267823  
Actual: 395000.0 Predicted: 472432.12324630097  
Actual: 485000.0 Predicted: 603610.7443359978  
Actual: 189000.0 Predicted: 265510.95064078644  
Actual: 230000.0 Predicted: 211426.0970951207  
Actual: 385000.0 Predicted: 450096.429882396  
Actual: 2000000.0 Predicted: 1167714.7009132989  
Actual: 285000.0 Predicted: 306105.30164395645  
Actual: 252700.0 Predicted: 235791.76072846726  
Actual: 329000.0 Predicted: 445202.12759767845  
Actual: 233000.0 Predicted: 182542.07781516388  
Actual: 937000.0 Predicted: 770772.6362713464  
Actual: 667000.0 Predicted: 720421.3644937165  
Actual: 438000.0 Predicted: 425453.26538001373  
Actual: 719000.0 Predicted: 632662.8267286904  
Actual: 580500.0 Predicted: 483124.37684689835  
Actual: 280000.0 Predicted: 394577.7351769097  
Actual: 687500.0 Predicted: 612005.0633891709  
Actual: 535000.0 Predicted: 546296.0233722813  
Actual: 322500.0 Predicted: 310154.85366116837  
Actual: 696000.0 Predicted: 758928.2687729485  
Actual: 550000.0 Predicted: 274964.72084580734  
Actual: 640000.0 Predicted: 902191.6805642731  
Actual: 240000.0 Predicted: 186135.2707384713  
Actual: 605000.0 Predicted: 510881.4283386357  
Actual: 625000.0 Predicted: 604942.8501490243  
Actual: 775000.0 Predicted: 468447.7656065114  
Actual: 861990.0 Predicted: 844932.0024205334  
Actual: 685000.0 Predicted: 501406.28249489143  
Actual: 309000.0 Predicted: 241314.35920821503  
Actual: 488000.0 Predicted: 771299.1363633759  
Actual: 210490.0 Predicted: 141382.57569085434  
Actual: 785000.0 Predicted: 744607.8386016972  
Actual: 450000.0 Predicted: 354250.2755314  
Actual: 1350000.0 Predicted: 1380308.5991872437  
Actual: 228000.0 Predicted: 243778.02010904625  
Actual: 345000.0 Predicted: 501283.56443797424  
Actual: 600000.0 Predicted: 433972.6850296147  
Actual: 585000.0 Predicted: 437030.7513319142  
Actual: 920000.0 Predicted: 875988.4195448048  
Actual: 885000.0 Predicted: 750144.4183226712  
Actual: 292500.0 Predicted: 302649.74342118576  
Actual: 301000.0 Predicted: 358233.73811112717  
Actual: 951000.0 Predicted: 1117505.0961307175  
Actual: 430000.0 Predicted: 423138.09907995537  
Actual: 650000.0 Predicted: 608980.1895598061  
Actual: 289000.0 Predicted: 265968.1027043946
```

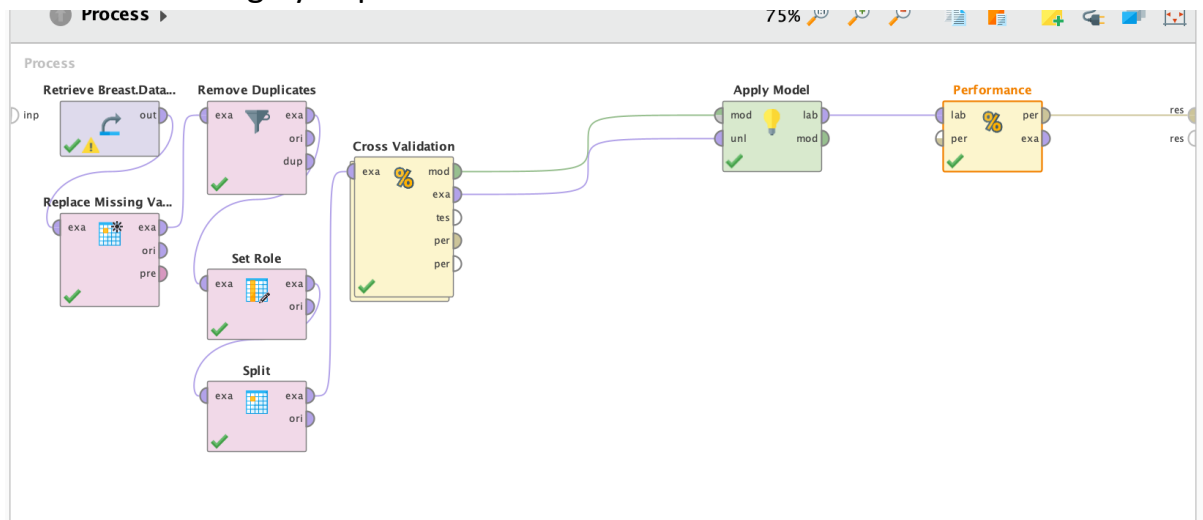

SettleRF:

```
[(.venv) sumitkafle@Sumits-MacBook-Air Lecture7Files % python3 SeattleRF.py  
[0.00282379 0.00661164 0.27767527 0.01411964 0.00191725 0.03179567  
0.01067447 0.00297048 0.30802346 0.02106374 0.00547352 0.0241069  
0.00194832 0.01302719 0.16316237 0.06965104 0.03159392 0.01336134]  
Actual: 221900.0 Predicted: 230480.56  
Actual: 538000.0 Predicted: 501990.243333333333  
Actual: 180000.0 Predicted: 219001.5  
Actual: 604000.0 Predicted: 563498.5  
Actual: 510000.0 Predicted: 512905.0  
Actual: 1230000.0 Predicted: 1266450.0  
Actual: 257500.0 Predicted: 267096.95  
Actual: 291850.0 Predicted: 253779.16666666667  
Actual: 229500.0 Predicted: 253933.5  
Actual: 323000.0 Predicted: 319138.67  
Actual: 662500.0 Predicted: 683966.78  
Actual: 468000.0 Predicted: 480815.15  
Actual: 310000.0 Predicted: 346906.75  
Actual: 400000.0 Predicted: 396178.6  
Actual: 530000.0 Predicted: 548849.0  
Actual: 650000.0 Predicted: 678295.52  
Actual: 395000.0 Predicted: 397644.5  
Actual: 485000.0 Predicted: 526281.0  
Actual: 189000.0 Predicted: 191467.4375  
Actual: 230000.0 Predicted: 226628.5  
Actual: 385000.0 Predicted: 400292.17  
Actual: 2000000.0 Predicted: 1657334.5  
Actual: 285000.0 Predicted: 300868.15  
Actual: 252700.0 Predicted: 228773.0  
Actual: 329000.0 Predicted: 334149.7  
Actual: 233000.0 Predicted: 222164.4  
Actual: 937000.0 Predicted: 865516.5  
Actual: 667000.0 Predicted: 654179.0  
Actual: 438000.0 Predicted: 443968.233333333334  
Actual: 719000.0 Predicted: 718580.45  
Actual: 580500.0 Predicted: 588289.0  
Actual: 280000.0 Predicted: 282718.66  
Actual: 687500.0 Predicted: 688211.18  
Actual: 535000.0 Predicted: 516813.5  
Actual: 322500.0 Predicted: 325139.76  
Actual: 696000.0 Predicted: 707574.0  
Actual: 550000.0 Predicted: 554708.9  
Actual: 640000.0 Predicted: 713552.32  
Actual: 240000.0 Predicted: 227339.595  
Actual: 605000.0 Predicted: 587712.88  
Actual: 625000.0 Predicted: 640828.0  
Actual: 775000.0 Predicted: 690012.9  
Actual: 861990.0 Predicted: 866072.26  
Actual: 685000.0 Predicted: 652912.5  
Actual: 309000.0 Predicted: 285987.49  
Actual: 488000.0 Predicted: 507464.0  
Actual: 210490.0 Predicted: 210233.1  
Actual: 785000.0 Predicted: 775627.46  
Actual: 450000.0 Predicted: 474995.49  
Actual: 1350000.0 Predicted: 1155583.0  
Actual: 228000.0 Predicted: 222064.5  
Actual: 345000.0 Predicted: 382619.5  
Actual: 600000.0 Predicted: 588490.3  
Actual: 585000.0 Predicted: 564674.88  
Actual: 920000.0 Predicted: 954170.0  
Actual: 885000.0 Predicted: 941498.0  
Actual: 292500.0 Predicted: 303918.5  
Actual: 301000.0 Predicted: 304667.26  
Actual: 951000.0 Predicted: 987594.68
```

Question 2(Solution)

Process

First the Breast Cancer dataset was retrieved from the RapidMiner local repository. The missing values were then imputed using mean imputation, that is, the average value of the column was used to fill in the missing values. The dataset was also cleaned and made reliable before modelling by duplicate data removal.



Training

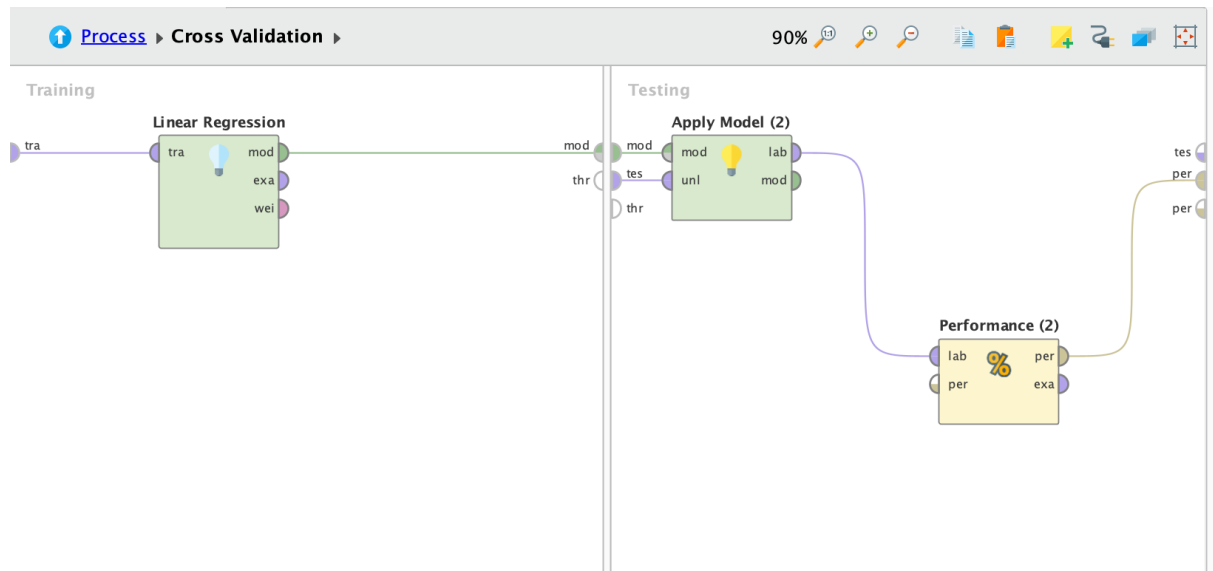
A Logistic Regression algorithm was used to train the classification model. The model was trained with 10-fold Cross Validation and 90% of data was used for training in each fold. The model was trained to predict the class for each record, either class 2 (Benign) or class 4 (Malignant), given the attributes.

Testing

The remaining 10% of the data in each fold was then used for testing the model. Predictions were made using the Apply Model operator and evaluations were made using the Performance operator. The overall accuracy of the model was 94.74% and precision for benign and malignant cases was 92.76% and 96.49% respectively. The recall rate for

benign and malignant class was 96.24% and 93.22% respectively, which shows high performance of the model in both the classes.

Here's the training and testing validation set performed in AI Altair Studio:



10% data used per fold for testing

Results:

Accuracy: 94.74%

Precision (2): 92.76%

Precision (4): 96.49%

Recall (2): 96.24%

Recall (4): 93.22%

PerformanceVector (Performance)

☒ Table View ☐ Plot View

accuracy: 94.65%

	true 2	true 4	class precision
pred. 2	205	16	92.76%
pred. 4	8	220	96.49%
class recall	96.24%	93.22%	

Conclusion:

Data was pulled and duplicates were removed, and missing data was filled in with mean imputation. Logistic Regression was trained and

tested with 10-fold Cross Validation, which resulted in 94.74% accuracy, high precision and good recall for benign and malignant class.